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RESEARCH ARTICLE



Acceptance, perceptions, and compliance for COVID-19 vaccines among students attending a rural university: An interventional study using brief video messages

Amber L. Harris Bozer, PhD^a , Subi Gandhi, PhD^b  and Dustin C. Edwards, PhD^c 

^aDepartment of Neuroscience, Tarleton State University, Stephenville, Texas, USA; ^bDepartment of Medical Lab Sciences, Public Health, and Nutrition Sciences, Tarleton State University, Stephenville, Texas, USA; ^cDepartment of Biological Sciences, Tarleton State University, Stephenville, Texas, USA

ABSTRACT

Objective: The purpose of this study was to examine the factors associated with vaccine compliance and the effectiveness of short-term video interventions on COVID-19 vaccine perceptions among students attending a state university located in rural Texas. **Participants:** A total of 298 students participated in an online survey. **Methods:** Students completed the COVID-19 Vaccine Acceptance Scale (COVID-VAC) and Perceptions of Vaccines Scale before and after watching one of three videos (neutral, educational, or disease effects). **Results:** Differences in vaccination status were observed for ethnicity and political leanings ($p < 0.05$). The video interventions did not impact COVID-VAC or Perceptions of Vaccines scores over time ($p > 0.05$). **Conclusions:** Short-term video interventions were ineffective in altering vaccine perceptions and improving acceptance of the COVID-19 vaccine in our study population. Impact of the type and duration of educational videos should be explored by future studies to combat vaccine hesitancy in future population-based studies.

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Introduction

Since Jan 1, 2020, there have been over one million deaths (302 per 100,000) in the United States due to COVID-19.¹ A substantial proportion of these fatalities could be attributed to the inadequate uptake of COVID-19 vaccines by the general population following their availability. Vaccines are among the most significant public health achievements of the twentieth century. They have eradicated smallpox, eliminated poliomyelitis in the Americas, and controlled other infectious diseases, including measles, diphtheria, and rubella.² Despite this, vaccines now face renewed criticism as segments of the public express dwindling confidence in their effectiveness.³ Just prior to the coronavirus pandemic, the World Health Organization declared vaccine hesitancy to be a great public health threat in controlling and preventing infectious diseases.⁴ Persistent vaccine hesitancy has limited the public health response to COVID-19 and hampered herd immunity efforts by immunization to reduce virus transmission at the community level.^{3,5,6} However, Morens and colleagues⁷ argue that achieving herd immunity may be unattainable due to the genetic instability of SARS-CoV-2, the virus responsible for COVID-19, which contrasts with the relative stability of poliovirus and measles virus.

Although the safety and efficacy of the Pfizer and Moderna COVID-19 vaccines have been repeatedly

demonstrated during clinical trials worldwide across all age groups, a significant portion of the American population remains hesitant to receive the vaccination.^{8,9} Complicating immunization efforts, non-pharmaceutical interventions such as state-mandated masking, social distancing, lockdowns, quarantines, and testing have caused societal polarization that extends to vaccine hesitancy and confidence, even though stark evidence shows unvaccinated individuals have higher incidence, hospitalization, and mortality rates due to COVID-19 compared to those who are vaccinated.^{6,10–15}

Several demographic and geopolitical factors have influenced the COVID-19 vaccination rates in the U.S. Although there has been significant acceptance and uptake of the COVID-19 vaccine by those aged 50 and over, other age groups lag in vaccination rates. One year after vaccine availability, 91% of those over the age of 65 and 82% of those aged 50–64 were fully vaccinated compared to those aged 18–49 (69%), 12–17 (60%), and 5–11 (30%).¹ Disparity in the uptake of vaccines was also observed among different racial/ethnic groups. In 2022, African Americans (57%) lagged significantly behind Whites (63%), Asians (85%), and Hispanics (65%) in receiving at least one dose of the COVID-19 vaccination across 38 states, and the overall vaccination rate has since plateaued for all racial and ethnic groups.^{1,16} Additional factors influencing the acceptance and

CONTACT Amber L. Harris Bozer  amberharrisbozer@gmail.com, amber.harris@go.tarleton.edu  Department of Neuroscience, Tarleton State University, Stephenville, TX, USA.

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adoption of the COVID-19 vaccine include lower socioeconomic status, limited educational attainment, rural residency, female gender, and affiliation with the Republican Party.^{17–20}

Geographical variations in vaccination rates have persisted within the United States. Gandhi and Harris Bozer (2020) studied rural college students' compliance with the flu vaccine. They found that despite having health insurance coverage, only 30% were vaccinated against the flu,²¹ in comparison to the national rates ranging from 8% to 39%.²² The primary factors contributing to low compliance were identified as the low perceived threat (20%) and time constraints due to busy schedules (11%).²¹ The vaccination rates for COVID-19 were much lower in rural areas compared to urban areas,²³ despite being administered free of charge during the pandemic.²⁴

Several studies have emerged regarding COVID-19 vaccination rates and hesitancy among college students during the pandemic. In a study conducted at a university ($n=457$) in Spring of 2021, among the unvaccinated students ($n=352$), nearly 53% indicated their intentionality to receive the COVID-19 vaccine when made available to the college body. Students who were healthcare workers and had a vaccinated family member were more likely to receive the COVID-19 vaccine. Non-vaccinated students who discussed vaccine information and had positive attitudes toward vaccination showed greater willingness to get the COVID-19 vaccine.²⁵ Another study conducted in a similar timeframe ($n=989$) demonstrated that perceived social norms significantly predicted vaccine hesitancy when accounting for demographics and pandemic experiences.²⁶ Other reasons noted in the literature for college students' hesitancy toward the COVID-19 vaccine included unknown side effects, skepticism about necessity, and mistrust toward the vaccine and the government, along with infodemic and political affiliation.^{26–29}

Purpose of the present study

When juxtaposed with their urban counterparts, individuals residing in rural regions encounter considerable health inequities attributable to both individual and environmental determinants.³⁰ This phenomenon can be readily extrapolated to the context of college students. According to the Health Resources and Services Administration, rural areas are defined as either micropolitan areas (with an urban core population of 10,000–49,999 individuals) or counties situated outside metropolitan areas (having an urban core of 50,000 or more individuals).^{31,32} This study examined college students' perceptions and attitudes toward vaccination during the study period in a rural country with a population of less than 45,000 residents.³³

Upon the availability of COVID-19 vaccines, vaccination rates were higher in urban areas than in rural areas.^{19,23} However, these prevalence studies were conducted in the general population.^{19,23,34,35} Recognizing approaches that can successfully enhance vaccine acceptance and lessen the effects of pandemics among university students is essential, considering that rural college students have different behaviors and health risk profiles compared to their counterparts,^{36–38} and generally possess a low perception of risk and

increased hesitancy regarding vaccine-preventable illnesses like the flu and COVID-19.^{21,28} Furthermore, high population density and close living conditions on college campuses facilitate outbreaks, as studies indicate a strong correlation between population density and COVID-19 transmission rates.³⁹

As college students are in the transition period between late adolescence and early adulthood, video-based interventions could aid in mitigating risk perception and hesitancy toward COVID-19 and promote responsible steps to reduce community spread in future pandemics. This premise is derived from the efficacy of establishing appropriate attitudes and intentions toward prevention behaviors, as demonstrated in a systematic review of 38 studies that highlighted the effectiveness of health promotion through video messaging,⁴⁰ as well as keeping the duration of the video in mind.⁴¹ Additionally, video messaging was an ideal intervention during the pandemic era, where COVID-19 had demonstrated high transmissibility rates at the community level.⁴²

Our study aimed to identify and elucidate perceptions of the COVID-19 vaccine among students attending a university in rural Texas utilizing the following research questions:

- a. What demographic factors differentiate individuals who are COVID-19 vaccine-compliant versus those who are non-compliant?
- b. What type of vaccine-related video information will positively change vaccine perceptions of COVID-19 (neutral, educational, flooding)?

Materials and methods

Participants

A University Institutional Review Board determined that the proposed study was exempt before participants were recruited for the study. Participants were recruited by sharing flyers, emails (using a mass email system), and various social media platforms during the Spring and Summer semesters of 2022. Participants were required to meet the following criteria: be at least 18 years of age, reside in the United States, be currently enrolled as a student, and provide informed consent before completing the anonymous survey.

Sampling and study design

We used convenience sampling and an experimental design. Participants were asked to complete survey questions in Google Forms related to "perception of vaccines" and were then randomized to one of the three interventional arms. Following the video presentation, each group was administered identical questions about their "perceptions of vaccines" they had previously answered before viewing the videos.

- i. The *control/neutral* group (arm 1) was exposed to a video unrelated to the vaccination topic (https://www.youtube.com/watch?v=I7STZsY_Ps; 2:22 length).

- ii. The first experimental group (arm 2) was exposed to an *educational* video produced by the Centers for Disease Control and Prevention (CDC) highlighting the advantages and importance of getting the COVID-19 vaccine (<https://www.youtube.com/watch?v=NeQ9ssmwG3w>; 1:38 length).
- iii. The last group (arm 3) was the experimental/flood-ing group (*disease effects*), which was exposed to a video showing complications of COVID-19 in a hos-pital setting. The video was produced by St. Charles Health (<https://youtu.be/Dz5ZVxWit3w>; 2:20 length). It demonstrated the pandemic's social justice and iso-lation aspects and depicted its burden on frontline workers, patients, and the healthcare system overall.

Measures

A digital survey was administered containing the informed consent, demographic, general vaccine, and monoclonal anti-body infusion questions (see Table 1 for a full list of survey questions and answer choices). Participants were only per-mitted by the software to have 1 response to the survey. Students may have received extra credit for survey comple-tion from their instructors (in good faith), but no other incentives or compensation occurred.

Instruments used in the study

- a. *COVID-19 Vaccine Acceptance Scale* (COVID-VAC): A six-item validated and reliable vaccination accep-tance scale was employed, encompassing constructs related to trust, risk, safety, and efficacy (see ques-tions in Table 1).⁴³ The items were measured on a Likert Scale (*Strongly agree* = 1 to *Strongly disagree* = 5), with higher scores indicating more vaccine accep-tance and lower indicating less vaccine acceptance. In the current study, the reliability of the scale was found to be high both before ($\alpha=0.91$, 95% CI 0.89–0.92) and after ($\alpha=0.91$, 95% CI 0.90–0.93) the administration of the video.
- b. *Perceptions of Vaccines Scale*: A nine-item validated and reliable scale was used to assess a participant's agreement with vaccines (see questions in Table 1),⁴⁴ containing three answer choices (1 = *agree* “yes,” 2 = *unsure*, and 3 = *disagree* “no.”) Higher scores on a single question indicate greater disagreement with that particular item. As with the COVID-VAC, this scale demonstrated high reliability before ($\alpha=0.88$, 95% CI 0.86–0.90) and after ($\alpha=0.90$, 95% CI 0.89–0.92) administration of the video.

Data analyses

Employing G*Power software, the optimal sample size for statistical analyses was computed for the largest statistical model (mixed ANOVA) a priori using 0.95 power

(probability of finding significance), keeping an alpha crite- rion level of 0.05 and an expected effect size of 0.25 ($n=66$).^{45–47} Data are presented as mean $\pm$ standard deviation. Frequency data with less than five participants in each cell were collapsed or excluded for analyses. Data were down- loaded into Microsoft Excel for cleaning, coding, and orga- nization, and SPSS and JASP software were used for statistical computations.^{48–50}

Demographic data and general questions were compared across video groups using Chi-square analyses with an alpha criterion of 0.05. The results include descriptive statistics for the number of children and care status. A Chi-square anal- ysis was run to compare the vaccination status of children and participants and to compare vaccine status by demo- graphic and general question variables.

To examine the influence of the short-term videos on COVID-VAC scores, we computed mixed ANOVAs using vaccine status (vaccinated/not vaccinated) and video group (neutral/education/disease effects) as between-subjects fac- tors and time (pre-survey/post-survey) as the within-subject factor. The Bonferroni post hoc test was employed solely in instances where there were main effects or interactions. Responses to the Perceptions of Vaccines questions were organized pre- and post-administration and segregated by video group and vaccination status.

Results

General questions and vaccine-related questions

A total of 360 participants responded to the survey. Four participants did not provide informed consent and were not presented with survey questions, and three were excluded for not completing the survey, leaving 353 for the final anal- ysis. A total of 55 more participants were excluded because they were not currently enrolled as students (alumni; $n=4$, former student/non-graduate; $n=1$, full-time administrators; $n=2$, full-time faculty; $n=5$, part-time faculty; $n=2$, staff; $n=11$, and other/not reported; $n=30$). However, analyses that include these 55 participants (not specific to students) can be viewed in Appendix B. After these exclusions, there were 298 remaining participants for analyses.

Comprehensive demographic comparisons are available in Table 2, with a breakdown in frequency and percentage for each demographic category by video group. Chi-square anal- yses to compare demographics by group indicated no differ- ences between groups on demographic categories other than political leanings (Table 2). Responses to the questions about first responder status, military status, university affiliation, and knowing someone hospitalized or died had too few responses in some cells to run analyses. The mean age of the sample was 28.28 ($SD=10.66$, $Range=19–69$), and par- ticipants were aged 18–24 years (55.03%), 25–40 years (29.87%), 40–55 years (12.42%), and 55+ years (2.68%). The study did not include an item related to student classifica- tions; however, the data revealed that 30% of students had completed high school, 27.52% had completed Associate's degrees, 33.89% had completed Bachelor's degrees, and 8.39% held professional degrees. Sample participants were White (75.17%), Hispanic or Latino (9.73%), Two or More

Table 1. Full survey questions with possible responses.

Demographics questions
Age (in years) open ended How do you describe your gender identity? Female Male Transgender man Transgender woman Gender non-conforming Intersex Other Prefer not to answer Ethnicity White Middle Eastern or North African Black or African American Native American or American Indian and/or Alaskan Native Asian Native Hawaiian or Pacific Islander Hispanic or Latino Two or more Other/unknown Ethnicity Hispanic or Latino Not Hispanic or Latino Education Some high school High school Associate's degree Trade school or vocational school Bachelor's degree Master's degree Professional or doctoral degree Marital status Married Married, but not cohabitating Not married, but cohabitating Not married or cohabitating Household income Less than \$25,000 \$25,000–\$50,000 \$50,000–\$85,000 \$85,000–\$140,000 More than \$140,000 Employment Employed, full-time Employed, part-time Retired Unemployed How many children do you have? (Choose 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or 10 or more) Please provide the ages of your children (check all that apply). Not applicable 4 years and under 5–11 years old 12–17 years old 18 years and older If you have children, are they under your care? Not applicable No, not under my care Yes, under my care part-time Yes, under my care full-time Political party Democrat Independent Libertarian Republican Green party Other What do you consider yourself mainly to be? (political ideology) Lightly conservative Moderately conservative Highly conservative Neutral Lightly liberal Moderately liberal

(Continued)

Table 1. Continued.

Demographics questions
Highly liberal
What is your affiliation with the university (check all that apply)?
Not affiliated
University student
Staff
Part-time faculty
Full-time faculty
Part-time administrator
Full-time administrator
If you are a college student, what is your residency status?
Not applicable
On-campus
Off-campus
If you are a college student, what is your classification?
Not applicable
Freshman
Sophomore
Junior
Senior
Post-baccalaureate student
Master's student
Doctoral student
Do you have medical insurance?
Yes
No
Are you a first responder?
No, I am not a first responder
Yes, I am an EMT
Yes, I am a paramedic
Yes, I am a firefighter or fire marshall
Yes, I am a police officer
Yes, I am a sheriff/sheriff's deputy
Yes, I am a correctional officer
Yes, I am park police or a park ranger
Yes, I am another kind of first responder
What is your military status?
I am not currently in the military and have never served
I am not currently in the military, but have served previously
I am currently in the military
General Vaccines Questions
Have you received the COVID-19 vaccine?
Yes, fully vaccinated plus boosted
Yes, fully vaccinated
Yes, partially vaccinated
No, I plan to
No, I do not plan to
If you are vaccinated, which one did you receive?
Not applicable – I am not vaccinated
Pfizer
Moderna
Johnson & Johnson
Have you tested positive for COVID-19?
No
Yes, while unvaccinated
Yes, while vaccinated
Yes, both while unvaccinated and later while vaccinated
If you said yes to the previous question, how did you learn that you had COVID-19?
Not applicable
From a home kit
From a medical provider
From symptoms alone
In the past, how many times did you test positive for COVID-19, with or without symptoms?
1
2
3 or more times
Not applicable
If you have children, have they received the COVID-19 vaccine?
Yes
No
Not applicable
What are your reasons for choosing to receive/not receive the COVID-19 vaccine?
Open ended
Please check all of the following reasons that led to you not receiving the COVID-19 vaccine. Please check all that apply.
Not applicable – I am vaccinated
Side effects

(Continued)

Table 1. Continued.

General Vaccines Questions
Time Mistrust of science Mistrust of vaccines Mistrust in this specific vaccine Concerns about the vaccine development Not knowing where to receive the vaccine Lack of knowledge about the risks of the vaccine I think the vaccine does not work The vaccine causes the disease itself My family does not want me to receive the vaccine My religious leader(s) do not want me to receive the vaccine I am concerned about the vaccine components I have a superior immune system and do not need vaccines Vaccines cause Autism Other reasons not listed What media modalities do you use for information about the COVID-19 vaccine? Please check all that apply Social interactions and chatting Television – network news Television – political shows Facebook Instagram LinkedIn Reddit Snapchat TikTok Twitter YouTube Scientific journal databases (PudMed, Scopus, OVID, Cochrane) Scholar research profiles (ResearchGate, ORCID, Acedemia.com, Google Scholar Profile) Other On a scale of 1–10, how likely are you to receive the COVID-19 vaccine in the future? (1–10) Please check all that apply. I have been hospitalized with COVID-19 in the past I know someone who was hospitalized from COVID-19 but did not die I know someone who was hospitalized from COVID-19 but did die I know someone who was never hospitalized from COVID-19 but did die I do not know anyone that has been hospitalized or died from COVID-19
Monoclonal Antibody Infusion Question
I would agree to receive a monoclonal antibody infusion Yes No Maybe
Perceptions of Vaccines Scale Questions
When new vaccines are made available to the general public, would you get vaccinated? Yes No Unsure Only people who are having underlying medical problems or who are pregnant should be vaccinated. Agree Disagree Unsure I will use vaccines if the government recommends them. Agree Disagree Unsure I will use vaccines if my doctor recommends it. Agree Disagree Unsure Vaccines will protect me from illness Agree Disagree Unsure I am concerned about the side effects of vaccines Agree Disagree Unsure I am concerned that vaccines have not been tested adequately Agree Disagree Unsure

(Continued)

Table 1. Continued.

Perceptions of Vaccines Scale Questions
Vaccines may lead to illness in some people
Agree
Disagree
Unsure
Vaccines will stop the spread of illness
Agree
Disagree
Unsure
COVID-VAC Scale Questions
COVID-19 is a dangerous health threat
Strongly disagree
Disagree
Unsure
Agree
Strongly agree
COVID-19 can be prevented by vaccination
Strongly disagree
Disagree
Unsure
Agree
Strongly agree
The risks of COVID-19 disease are greater than risks of the vaccine
Strongly disagree
Disagree
Unsure
Agree
Strongly agree
The COVID-19 vaccines available to me are safe
Strongly disagree
Disagree
Unsure
Agree
Strongly agree
I trust that my government is able to deliver the COVID-19 vaccine to everyone, everywhere in my country, equally
Strongly disagree
Disagree
Unsure
Agree
Strongly agree
I trust the science behind the COVID-19 vaccines
Strongly disagree
Disagree
Unsure
Agree
Strongly agree

(7.05%), Black or African American (5.37%), Asian (1.34%) Native American or American Indian, and/or Alaskan Native (1.01%), and Native Hawaiian or Pacific Islander (.34%). The sample was primarily not Hispanic or Latino (79.87%). Females were the primary respondents (72.82%), with the remaining participants identifying as male (22.15%), Transgender man (.34%), gender non-conforming (2.68%), and prefer not to answer (2.01%).

With regards to parental status, a total of 210 (70.47%) participants did not have children, 24 (8.05%) participants had one child, 38 (12.75%) participants had 2 children, 14 (4.70%), participants had 3 children, 8 (2.69%) participants had 4 children, and 4 (1.34%) participants had 5 or more children. Of the participants with children, 11 reported that the children were not under their care, 3 reported part-time care, and 68 reported full-time care. A Chi-square comparison of child vaccination status by participant vaccination status indicated that participants who received the COVID-19 vaccine were significantly more likely to vaccinate their children, $\chi^2(1) = 27.16$, $p < 0.001$. Specifically, tabulations were as follows: vaccinated participants with children who had

received the vaccine ($n = 34$), vaccinated participants with children who had not received the vaccine ($n = 24$), unvaccinated participants with children who had received the vaccine ($n = 1$), and unvaccinated participants with children who had not received the vaccine ($n = 29$).

To assess the factors that differed between COVID-19 vaccine-compliant and non-compliant, Chi-square analyses were conducted to compare demographics and general variables by vaccination status. The analyses revealed no differences in most demographics ($p > 0.05$). However, there were differences in ethnicity, political party, and political ideology across vaccination status (Table 3).

A large number of participants reported not being vaccinated ($n = 97$, 32.5% of the total sample). Vaccinated participants ($n = 201$) reported receiving Moderna ($n = 116$; 57%), Pfizer ($n = 73$; 36%), and Johnson & Johnson Vaccines ($n = 9$; 4%). Three participants were unsure about the type of vaccine they received. Vaccinated participants reported not testing positive in the past ($n = 105$; 52%) and testing positive ($n = 43$ while unvaccinated; 21%, $n = 40$ while vaccinated; 20%, and $n = 13$ while unvaccinated and later while

Table 2. Chi square comparisons of demographic data and general questions by video group.

Variable	Sample (N=298)		Neutral group (n=97)		Education group (n=128)		Effects group (n=73)		χ^2	p
	n	%	n	% of group	n	% of group	n	% of group		
Age									1.84	0.934
18–24 years	164	55.03%	55	56.70%	72	56.25%	37	50.68%		
25–40 years	89	29.87%	28	28.87%	38	29.69%	23	31.51%		
40–55 years	37	12.42%	11	11.34%	16	12.50%	10	13.70%		
55+	8	2.68%	3	3.09%	2	1.56%	3	4.11%		
Ethnicity									3.34	0.911
Black or African Am.	16	5.37%	5	5.15%	5	3.91%	6	8.22%		
Hispanic or Latino	29	9.73%	10	10.31%	14	10.94%	5	6.85%		
Two or more	21	7.05%	6	6.19%	9	7.03%	6	8.22%		
White	224	75.17%	73	75.26%	96	75.00%	55	75.34%		
Collapsed category*	8	2.69%	3	3.09%	4	3.13%	1	1.37%		
Asian*	4	1.34%	1	1.03%	2	1.56%	1	1.37%		
Middle Eastern or North African*	0	0.00%	0	0.00%	0	0.00%	0	0.00%		
Nat. Am. or Am. Indian and/or Alaskan Nat.*	3	1.01%	2	2.06%	1	0.78%	0	0.00%		
Nat. Hawaiian or Pacific Islander*	1	0.34%	0	0.00%	1	0.78%	0	0.00%		
Ethnicity									6.79	0.147
Hispanic or Latino	54	18.12%	17	17.53%	29	22.66%	8	10.96%		
Not Hispanic or Latino	238	79.87%	77	79.38%	96	75.00%	65	89.04%		
Other^	6	2.01%	3	3.09%	3	2.34%	0	0.00%		
Gender									2.36	0.671
Female	217	72.82%	70	72.16%	91	71.09%	56	76.71%		
Male	66	22.15%	20	20.62%	31	24.22%	15	20.55%		
Collapsed category*	15	5.03%	7	7.22%	6	4.69%	2	2.74%		
Transgender man*	1	0.34%	1	1.03%	0	0.00%	0	0.00%		
Gender non-conforming*	8	2.68%	6	6.19%	2	1.56%	0	0.00%		
Prefer not to answer	6	2.01%	0	0.00%	4	3.13%	2	2.74%		
Education									9.31	0.157
Collapsed category*	90	30.20%	30	30.93%	43	33.59%	17	23.29%		
Some high school*	2	0.67%	2	2.06%	0	0.00%	0	0.00%		
High school diploma*	84	28.19%	27	27.84%	41	32.03%	16	21.92%		
Trade school or vocational school*	4	1.34%	1	1.03%	2	1.56%	1	1.37%		
Associate's degree	82	27.52%	31	31.96%	33	25.78%	18	24.66%		
Bachelor's degree	101	33.89%	33	34.02%	38	29.69%	30	41.10%		
Collapsed category*	25	8.39%	3	3.09%	14	10.94%	8	10.96%		
Master's degree**	23	7.72%	3	3.09%	12	9.38%	8	10.96%		
Professional or doctoral degree**	2	0.67%	0	0.00%	2	1.56%	0	0.00%		
Marital status									0.41	0.816
Collapsed category* No	159	53.36%	50	51.55%	71	55.47%	38	52.05%		
Not married or cohabitating*	158	53.02%	50	51.55%	71	55.47%	37	50.69%		
Married, but not cohabitating*	1	0.34%	0	0.00%	0	0.00%	1	1.37%		
Collapsed category*Yes	139	46.64%	47	48.45%	57	44.53%	35	47.95%		
Married*	92	30.87%	28	28.87%	39	30.47%	25	34.25%		
Not married, but cohabitating	47	15.77%	19	19.59%	18	14.06%	10	13.70%		
Household income									6.84	0.554
Less than \$25,000	110	36.91%	36	37.11%	48	37.50%	26	35.62%		
\$25,000–\$50,000	63	21.14%	22	22.68%	20	15.63%	21	28.77%		
\$50,000–\$85,000	48	16.11%	14	14.43%	25	19.53%	9	12.33%		
\$85,000–\$140,000	52	17.45%	17	17.53%	25	19.53%	10	13.70%		
More than \$140,000	25	8.39%	8	8.25%	10	7.81%	7	9.59%		
Employment									4.29	0.637
Employed, full time	109	36.58%	32	32.99%	44	34.38%	33	45.21%		
Employed, part time	106	35.57%	38	39.18%	46	35.94%	22	30.14%		
Retired	6	2.01%	3	3.09%	2	1.56%	1	1.37%		
Unemployed	77	25.84%	24	24.74%	36	28.13%	17	23.29%		
Political party									19.9	#0.011
Democrat	64	21.48%	20	20.62%	29	22.66%	15	20.55%		
Independent	59	19.80%	31	31.96%	17	13.28%	11	15.07%		
Libertarian	17	5.71%	2	2.06%	12	9.38%	3	4.11%		
Republican	104	34.90%	28	28.87%	44	34.38%	32	43.84%		
Other	54	18.12%	16	16.49%	26	20.31%	12	16.44%		
Political ideology									17.6	0.127
Lightly conservative	37	12.42%	9	9.28%	17	13.28%	11	15.07%		
Moderately conservative	72	24.16%	22	22.68%	34	26.56%	16	21.92%		
Highly conservative	25	8.39%	5	5.16%	9	7.03%	11	15.07%		
Neutral	58	19.46%	17	17.53%	29	22.66%	12	16.44%		

(Continued)

Table 2. Continued.

Variable	Sample (N = 298)		Neutral group (n = 97)		Education group (n = 128)		Effects group (n = 73)		χ^2	p
	n	%	n	% of group	n	% of group	n	% of group		
Lightly liberal	26	8.73%	8	8.25%	14	10.94%	4	5.48%		
Moderately liberal	45	15.10%	18	18.56%	15	11.72%	12	16.44%		
Highly liberal	35	11.75%	18	18.56%	10	7.81%	7	9.59%		
Medical insurance status										
Insured	251	84.23%	77	79.38%	108	84.38%	66	90.41%	3.82	0.148
Not insured	47	15.77%	20	20.62%	20	15.63%	7	9.59%		

*Collapsed within analysis due to small numbers per cell: #statistically significant at 0.05.

Table 3. Participant data by vaccination status.

Variable	Sample (N = 298)		Not vaccinated (n = 97)		Vaccinated (n = 201)		χ^2	p
	n	%	n	%	n	%		
Age							1.99	0.575
18–24 years	164	55.03%	52	53.61%	112	55.72%		
25–40 years	89	29.87%	32	32.99%	57	28.36%		
40–55 years	37	12.42%	12	12.37%	25	12.44%		
55+	8	2.69%	1	1.03%	7	3.48%		
Ethnicity							14.27	.006*
Black or African Am.	16	5.37%	2	2.06%	14	6.97%		
Hispanic or Latino	29	9.73%	4	4.12%	25	12.44%		
Two or more	21	7.05%	4	4.12%	17	8.46%		
White	224	75.17%	86	88.66%	138	68.66%		
Collapsed category*	8	2.69%	1	1.03%	7	3.48%		
Ethnicity							5.07	0.080
Hispanic or Latino	54	18.12%	11	11.34%	43	21.39%		
Not Hispanic or Latino	238	79.87%	83	85.57%	155	77.11%		
Other^	6	2.01%	3	3.09%	3	1.49%		
Gender							5.73	0.057
Female	217	72.82%	67	69.07%	150	74.63%		
Male	66	22.15%	28	28.87%	38	18.91%		
Collapsed category*	15	5.03%	2	2.06%	13	6.47%		
Education							3.44	0.328
Collapsed category (high school or trade school)	90	30.20%	30	30.93%	60	29.85%		
Associate's degree	82	27.52%	28	28.87%	54	26.87%		
Bachelor's degree	101	33.89%	35	36.08%	66	32.84%		
Collapsed category (professional degree)	25	8.39%	4	4.12%	21	10.45%		
Marital status							1.39	0.239
Collapsed category (not married, not cohabitating)	159	53.36%	47	48.45%	112	55.72%		
Collapsed category (married, cohabitating)	139	46.64%	50	51.55%	89	44.28%		
Household income							7.79	0.100
Less than \$25,000	110	36.91%	27	27.84%	83	41.29%		
\$25,000–\$50,000	63	21.14%	27	27.84%	36	17.91%		
\$50,000–\$85,000	48	16.11%	15	15.46%	33	16.42%		
\$85,000–\$140,000	52	17.45%	17	17.53%	35	17.41%		
More than \$140,000	25	8.39%	11	11.34%	14	6.97%		
Employment							1.87	0.600
Employed, full time	109	36.58%	39	40.21%	70	34.83%		
Employed, part time	106	35.57%	32	32.99%	74	36.82%		
Retired	6	2.01%	3	3.09%	3	1.49%		
Unemployed	77	25.84%	23	23.71%	54	26.87%		
Political party							76.36	<.001*
Democrat	64	21.48%	1	1.03%	63	31.34%		
Independent	59	19.80%	14	14.43%	45	22.39%		
Libertarian	17	5.70%	8	8.25%	9	4.48%		
Republican	104	34.90%	64	65.98%	40	19.90%		
Other	54	18.12%	10	10.31%	44	21.89%		
Political ideology							113.91	<.001*
Lightly conservative	37	12.42%	10	10.31%	27	13.43%		
Moderately conservative	72	24.16%	43	44.33%	29	14.43%		
Highly conservative	25	8.39%	24	24.74%	1	0.50%		
Neutral	58	19.46%	17	17.53%	41	20.40%		
Lightly liberal	26	8.73%	2	2.06%	24	11.94%		
Moderately liberal	45	15.10%	1	1.03%	44	21.89%		
Highly liberal	35	11.75%	0	0.00%	35	17.41%		
Medical insurance status							0.333	0.564
Insured	251	84.23%	80	82.47%	171	85.08%		
Not insured	47	15.77%	17	17.53%	30	14.93%		

vaccinated; 6%). Unvaccinated participants reported not testing positive ($n=37$; 38%) and testing positive ($n=60$; 62%). Participants reported learning that they had contracted COVID-19 from a home kit ($n=27$; 9%), from a medical provider ($n=108$; 36%), and from symptoms alone ($n=20$; 7%). Participants reported testing positive once ($n=120$; 40%), twice ($n=34$; 11%), or three or more times ($n=7$; 2%).

Participants were asked to rate their likelihood of receiving the COVID-19 vaccine using a 10-point Likert scale. The mean for vaccinated participants was 8.76 ($SD=2.69$), while the mean for unvaccinated participants was 1.57 ($SD=1.26$). Vaccinated participants were more likely to accept the monoclonal antibody infusion in the future ($p<0.001$). Among the vaccinated participants, 48% ($n=97$) reported that they were willing to receive the infusion, 6% ($n=12$) stated that they were not willing, and 46% ($n=92$) were unsure or maybe willing. In contrast, among the unvaccinated participants, 11% ($n=11$) expressed willingness to receive the infusion, 49% ($n=48$) were not willing, and 39% ($n=38$) were unsure or maybe willing.

To assess reasons for vaccine uptake, open-ended data collected on reasons for not receiving the vaccine and media modalities for obtaining information about the vaccine are available in Appendix A. The top reason selected for not receiving the vaccine was mistrust in this specific vaccine ($n=75$), followed by concerns about the vaccine development ($n=70$), potential side effects ($n=68$), doubts about the vaccine's effectiveness ($n=53$), worries about the vaccine components ($n=45$), time constraints ($n=29$), lack of knowledge about the vaccine risks ($n=28$), other unlisted reasons ($n=27$), mistrust of science ($n=23$), mistrust of vaccines ($n=20$), belief that the vaccine causes the disease itself ($n=19$), a belief in one's superior immune system that makes vaccines unnecessary ($n=16$), the belief that vaccines cause autism ($n=10$), family opposition to vaccination ($n=8$), and difficulty finding a location to receive the vaccine ($n=1$).

Vaccine acceptance scales

To assess which video type could positively enhance vaccine perceptions, vaccine acceptance scales were administered (COVID-VAC and Perceptions of Vaccines Scale). The mixed ANOVA results for comparing COVID-VAC scores, using vaccine status and video group as the between-subjects factors and time as the within-subject factor, are presented in Table 4. The absence of three-way interactions ($p>0.05$) suggests that none of the short-term videos effectively increased vaccine acceptance. Table 5 provides frequency tabulations of perceptions of vaccine questions, categorized by video group and vaccination status, at different intervals over time.

Discussion

Main findings of this study

The primary objectives of this study were to (a) identify the demographic factors associated with COVID-19 vaccine compliance in a sample of students attending a rural

university and (b) determine what type of vaccine-related video information would positively change vaccine perceptions. College students often participate in risky behaviors and often make unhealthy lifestyle decisions,^{36,51} so investigating factors that may contribute to vaccine hesitancy in this group is of interest. The first research objective was to assess the demographic factors associated with vaccine compliance, and an assessment of factors that differ between vaccinated and unvaccinated groups was conducted. Findings indicated no significant differences in vaccination status across most demographics or insurance status; however, differences across vaccination status were observed for ethnicity, political party, and political ideology, as corroborated by the recent literature.^{40,52,53} Approximately 45% of our sample was composed of participants who identified as conservative. A 2022 study supports our finding that more conservative political ideologies in rural areas contribute to vaccine hesitancy and that tailored interventions that specifically appeal to the interests of these groups (and address common misconceptions) are an avenue to be explored to facilitate overcoming negative perceptions of the vaccine.^{54–56} These findings suggest selective factors may influence vaccination status, a finding that merits further investigation in samples of students attending a rural university.

When factors associated with vaccine hesitancy are discovered, encouraging health literacy is a solution. Health literacy refers to the capacity to seek out, comprehend, assess, and use health information from diverse communication channels. It plays a crucial role in recognizing misinformation, including fake news.⁵⁷ Montagni and colleagues found that individuals with a limited ability to distinguish fake news are more likely to be hesitant to receive the COVID-19 vaccine and educational and political interventions are suggested.⁵⁷ Similarly, vaccine literacy warrants consideration when examining the relationships between vaccines and their uptake.⁵⁸

Among the participants in this study, 97 individuals (32.5% of the total sample) reported not being vaccinated against COVID-19. Additionally, a substantial portion (70%) of the participants in the study did not have children. However, the findings revealed a crucial relationship between COVID-19 vaccination and the vaccination status of participants' children. Specifically, those who had been vaccinated against COVID-19 were more likely to have vaccinated their children as well. This trend may suggest that a parent's vaccination status could influence the vaccination of their children.⁵⁹

Pertaining to the second research objective to determine what type of vaccine-related video information would positively change vaccine perceptions, evaluations of video interventions were conducted. Previous literature has indicated that short-term media interventions (using platforms such as YouTube and TikTok) degrading vaccines can negatively impact vaccine perceptions and acceptance/uptake.^{60,61} The current study data indicate that short-term media educational video interventions were not enough to positively enhance vaccine acceptance/uptake in this sample of

Table 4. COVID-19 vaccine acceptance survey results by video group and vaccination status.

Question	No vaccine			Vaccine			Main effects			Interactions					
	Mean (pre)	Standard deviation (pre)	Mean (post)	Standard deviation (post)	Mean (pre)	Standard deviation (pre)	Mean (post)	Standard deviation (post)	Status	Group	Time	Status* group	Status* time	Time* group	3-way interaction
COVID-19 is a dangerous health threat															
Neutral video group	3.36	1.20	3.45	1.15	1.62	0.91	1.71	1.06	<.001**	0.793	0.577	0.989	.017*	0.18	0.237
Education video group	3.29	1.04	3.33	1.02	1.71	0.99	1.52	0.82							
Effects video group	3.29	1.27	3.52	1.29	1.77	0.98	1.64	0.91							
COVID-19 can be prevented by vaccination															
Neutral video group	4.23	0.76	4.26	0.73	2.29	1.16	2.12	1.14	<.001**	0.467	0.027*	0.868	0.034*	0.822	0.983
Education video group	4.24	0.77	4.20	0.82	2.24	1.19	2.02	1.13							
Effects video group	4.33	0.80	4.33	0.73	2.52	1.26	2.31	1.25							
The risks of COVID-19 disease are greater than the risks of the vaccine															
Neutral video group	3.94	1.06	3.84	1.04	1.41	0.68	1.47	0.66	<.001**	0.217	0.051	0.243	0.293	0.195	0.398
Education video group	3.58	1.06	3.58	1.03	1.59	1.06	1.53	0.95							
Effects video group	4.05	0.87	3.76	0.94	1.79	1.05	1.67	1.02							
The COVID-19 vaccines available to me are safe															
Neutral video group	3.90	0.79	4.00	0.82	1.70	0.80	1.85	0.90	<.001**	0.162	0.286	0.915	0.828	0.229	0.351
Education video group	3.82	1.05	3.91	0.97	1.72	0.90	1.66	0.79							
Effects video group	4.10	0.94	4.05	1.07	2.00	0.91	2.00	0.84							
I trust that my government is able to deliver the COVID-19 vaccine to everyone everywhere in my country, equally															
Neutral video group	3.61	1.28	3.58	1.29	3.00	1.34	2.92	1.29	<.001**	0.339	<.001**	0.725	0.774	0.455	0.882
Education video group	3.47	1.38	3.36	1.33	2.74	1.14	2.59	1.16							
Effects video group	3.52	1.33	3.38	1.32	3.08	1.22	2.96	1.20							
I trust the science behind the COVID-19 vaccines															
Neutral video group	4.26	0.82	4.36	0.76	1.67	0.87	1.65	0.81	<.001**	0.081	0.111	0.468	0.231	0.030*	0.725
Education video group	4.22	0.93	4.09	1.04	1.77	1.00	1.63	0.93							
Effects video group	4.38	0.81	4.38	0.81	2.15	1.11	2.06	0.98							

Note. *Statistically significant at $p < .05$. **Statistically significant at $p < .001$. Survey items 1–6, based on responses ranging from strongly agree = 1 to strongly disagree = 5.

Table 5. Perceptions of vaccines split by group and vaccination status.

Question	Neutral group (n=97)			Education group (n=128)			Effects group (n=73)		
	Not vaccinated (n=31)		Vaccinated (n=66)	Not vaccinated (n=45)		Vaccinated (n=83)	Not vaccinated (n=21)		Vaccinated (n=52)
	Pre	n/post n		Pre	n/post n		Pre	n/post n	
When new vaccines are made available to the general public, would you get vaccinated?									
Agree	0/0		45/46	1/0		56/63	1/1		30/37
Unsure	11/12		17/16	23/19		25/19	10/7		19/12
Disagree	20/19		4/4	21/26		2/1	10/13		3/3
Only people who are having underlying medical problems or who are pregnant should be vaccinated.									
Agree	6/7		2/4	3/2		1/1	1/1		2/1
Unsure	7/4		3/2	12/9		11/8	6/3		4/3
Disagree	18/20		61/60	30/34		71/74	14/17		46/48
I will use vaccines if the government recommends them.									
Agree	0/0		33/32	0/0		37/42	0/0		22/23
Unsure	5/7		22/23	6/6		32/27	3/3		24/24
Disagree	26/24		11/11	39/39		14/14	18/18		6/5
I will use vaccines if my doctor recommends it.									
Agree	3/4		57/58	10/10		74/75	1-2		43/45
Unsure	9/11		9/7	15/14		8/6	9/7		9/6
Disagree	19/16		0/1	20/21		1/2	11/12		0/1
Vaccines will protect me from illness									
Agree	11/8		58/56	20/15		64/71	5/4		45/46
Unsure	4/11		5/8	13/15		11/6	5/6		3/1
Disagree	16/12		3/2	12/15		8/6	11/11		4/5
I am concerned about the side effects of vaccines									
Agree	27/26		24/21	32/38		38/30	17/20		23/23
Unsure	1/2		9/11	4/0		15/12	2/1		11/6
Disagree	3/3		33/34	9/7		30/41	2/0		18/23
I am concerned that vaccines have not been tested adequately									
Agree	29/27		18/18	36/36		15/14	18/20		18/13
Unsure	1/4		8/4	4/4		21/18	2/0		7/13
Disagree	1/0		40/44	5/5		47/51	1-1		27/26
Vaccines may lead to illness in some people									
Agree	29/28		40/39	38/33		54/53	18/17		29/29
Unsure	2/3		12/11	6-7		17/7	2-3		8-6
Disagree	0/0		14/16	1-5		12/23	1-1		15/17
Vaccines will stop the spread of illness									
Agree	7/6		50/53	16/13		61/65	4/4		38/42
Unsure	7/9		8/6	11/15		10/8	3/7		5/6
Disagree	17/16		8/7	18/17		12/10	14/10		9/4

students attending a rural university, as evidenced by the lack of interaction between status, group, and time in the comparison of COVID-VAC scores and the lack of change over time in the frequency tabulations of the Perceptions of Vaccines Scale.

What is already known

Approximately 60 million people in the U.S. reside in rural counties, constituting approximately 20% of the population.²³ During the pandemic, disparities in vaccination rates began to emerge, with rural residents exhibiting lower vaccination rates since the beginning of 2021.¹⁹ Since January 2021, youth and adults residing in rural areas exhibited lower vaccination rates compared to their urban and suburban counterparts.⁶² Furthermore, the 2021 CDC report uncovered that 75.4% of the U.S. urban population (≥ 5 years old) received one dose of the Janssen vaccine or two doses of Pfizer-BioNTech or Moderna vaccine compared to only 58.5% of the U.S. rural population.¹⁹ On the other hand, the number of weekly COVID-19 cases per 100,000 population in rural areas began to surpass that of urban areas in October 2020.⁶³ This inverse relationship between the low vaccination rates and increasing case rates could potentially create a critical situation, leading to elevated morbidity and mortality rates associated with COVID-19 among rural populations. The literature suggests that rural dwellers are more likely to have comorbidities, are older on average, and are less likely to have health insurance.²³ As such, disparities in the COVID-19 vaccine uptake between urban and rural areas could hinder progress in ending the pandemic and future pandemics.^{8,19,23,64} This further highlights the importance of prioritizing mitigation efforts in both geographic areas during large-scale outbreaks.

A national survey conducted during the coronavirus pandemic, but before vaccines for COVID-19 were available, showed that approximately 3 in 10 adults were unsure about taking the vaccine and 1 in 10 did not intend to be vaccinated, indicating that targeted and multipronged efforts would be needed to increase vaccination rates.⁶⁵ The literature reveals conflicting evidence regarding the impact of educational videos and blogs on vaccine-related attitudes. A prior research examining (HPV) vaccine perceptions and behaviors among East Coast University in the US revealed that individuals exposed to negative blog content viewed the vaccine as less reliable, developed more unfavorable opinions about it, and were less likely to get vaccinated. Conversely, exposure to the positive blog did not change vaccine-related risk perceptions, attitudes, or intentions.⁶⁶ This indicates that negative details regarding vaccine effectiveness and safety have a larger influence on the vaccine adoption habits of university students than positive ones.⁶⁶ In contrast, a different investigation revealed that Southern US university students exhibited higher levels of knowledge, perception, and intentions regarding both video and text-based information for the meningitis B vaccine.⁶⁷ Although vaccine attitudes were studied using video-based interventions during the pandemic involving a different study population,⁶⁸ to our knowledge, this is the first

interventional study conducted among college students during the pandemic to assess COVID-19 vaccine uptake attitudes with video messaging intervention.

Limitations

Limitations of the current study include the potential for information bias due to the self-report design of the study and the inability to yield nuanced insights due to the inadequate number of responses received for variables such as first responder status, military status, and university affiliation. Since the study survey was deployed *via* the internet or in a digital format *via* social media, we may have excluded some individuals, especially rural older adults with limited digital literacy. Only 3% of our study sample were over the age of 55 years, which reduces the likelihood of age affecting the study results significantly.⁶⁹ The convenience sampling design of the study may have compromised the generalizability of our findings to a certain extent, along with the classification (undergraduate versus graduate students), as we lacked an item related to this questionnaire that would allow for comparisons with other vaccine hesitancy profiles examined at different universities. Furthermore, like other work investigating vaccine hesitancy in rural samples,⁴⁰ our study sample comprised a slightly higher proportion of white students (75.17%). Our sample also included Hispanic (9.73%), Black (5.37%), and Asian (1%) students. The current university's demographic composition consists of 64% white, 23% Hispanic or Latino, 8% Black students, and 1.34% Asian students. The mean age of the sample was 28 years, and while about 55% of participants were aged 18–24 years (55.03%), there were older students in the sample. This may not reflect the age distribution of all universities.

Future research

Future research could seek to unravel the findings related to significant differences in ethnicity, political party, and political ideology across vaccinated and unvaccinated students. Elucidating the contingencies that may influence vaccination status merits further investigation. More comprehensive studies should be conducted to ascertain the nature and length of educational videos for creating health promotion campaigns aimed at addressing vaccine hesitancy, particularly among adolescents and young adults.

Conclusion

Our analysis revealed that short informative videos did not improve rural university student attitudes toward COVID-19 vaccine acceptance, highlighting a need for more targeted and extended interventions to increase compliance with novel vaccines. Some key factors influencing vaccination, such as ethnicity, political party, and ideology, need further validation through other robust studies among college students. Factors such as these and others could be uncovered for targeted interventions to decrease vaccine hesitancy during public health emergencies such as the COVID-19 pandemic.

Conflict of interest disclosure

The authors have no conflicts of interest to report. The authors confirm that the research presented in this article met the ethical guidelines, including adherence to the legal requirements, of the USA and received approval from the Institutional Review Board of Tarleton State University.

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ORCID

Amber L. Harris Bozer  <http://orcid.org/0000-0003-0487-5237>

Subi Gandhi  <http://orcid.org/0000-0003-0269-2082>

Dustin C. Edwards  <http://orcid.org/0000-0001-6409-8041>

Data availability statement

Data will be made available on Open Science Framework.

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